

# SIHAN JIANG

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## EDUCATION

### Yale University

08/2025-Present

*M.S. in Biostatistics (Data Science Pathway)*

New Haven, CT, U.S.

- **GPA:** 3.89/4.00
- **Relevant Coursework:** Large Language Models, Intermediate Machine Learning, Biomedical Data Science: Mining and Modeling, Linear Models

### Duke Kunshan University (DKU) & Duke University

08/2021-05/2025

*B.S. in Applied Mathematics and Computational Science/Computer Science, by DKU*

Kunshan, Jiangsu, China

*B.S. in Interdisciplinary Studies (Computer Science; Applied Math & Computational Science), by Duke*

Durham, NC, U.S.

- **GPA:** 3.81/4.00
- **Honors & Recognition:** Latin Honors (Cum Laude); Dean's List (Fall 2021, Spring 2022, Spring 2024)
- **Relevant Coursework:** Principles of Machine Learning, Design & Analysis of Algorithms, Cell and Molecular Biology, Integrated Science - Chemistry

## PUBLICATIONS & CONFERENCE PRESENTATIONS

- Liu, T., Wang, A. X., Panescu, A., Chen, L. X., Long, W., Wei, X., Jing, Y., Zeng, Z., Chen, J., **Jiang, S.**, Wang, Z., Gu, S., Chen, S., Hu, X., Shao, H., Xu, L., Zheng, W., Cao, Z., Fang, A., Yu, B., Sun, O., Ying, R., Cohan, A., Chen, Q., Xue, L., Ding, K., Du, Y., Jin, W., Yang, Z., Zitnik, M., Zou, J., Xu, H., & Zhao, H. (2026). Benchmarking AI Agents for Addressing Scientific Challenges Across Scales. *Manuscript submitted to Nature. (Task Contributor; Equal Contribution)*
- Liu, T., **Jiang, S.**, Zhang, F., Sun, K., Head-Gordon, T., & Zhao, H. (2026). DrugPlayGround: Benchmarking Large Language Models and Embeddings for Drug Discovery. *Manuscript under review by npj Drug Discovery. (Co-first Authors)*
- **Jiang, S.**, & Huang, M. C. (2025). Diabetes Prevention Chatbots to Help Dietary Self-management for the Chinese Population. *2025 Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, 1–7. <https://doi.org/10.1109/EMBC58623.2025.11254311> (*First Author*)
- **Jiang, S.**, Shen, J., & Chen, M. (2024). Factors Included in AI Algorithms Assisting Precision Medicine in Cancer Treatment: A Systematic Review. Poster Presentation. *2024 Consortium of Universities for Global Health (CUGH) Conference. (Co-first Authors; Presenter)*

## RESEARCH EXPERIENCE

### Benchmarking AI Agents for Addressing Multi-Scale Scientific Challenges

01/2026-Present

*Research Assistant (Supervisor: Prof. Hongyu Zhao, Yale University)*

New Haven, CT, U.S.

- Designed drug safety and pharmacological evaluation tasks for AI agents, including toxicophoric detection, hERG prediction, PAINS false positive filter and classification, metabolic soft spot identification, and “cyanide trap” detection; further tested their ability to detect dataset corruption and instruction conflicts in research settings.
- Developed task-based evaluation metrics to systematically assess scientific validity, reasoning reliability, and failure behaviors, including F1 score, multi-tier toxicity alert accuracy, and stratification accuracy scores.
- Investigated failure modes across general LLMs, multidomain agents, and chemistry-specialized agents, covering schema mismatch, incorrect method selection, tool-invocation errors, runtime exceptions, and objective misinterpretation.
- Built reproducible analysis and visualization pipelines for large-scale comparative model evaluation.

### End-to-End Nextflow Data Processing Pipeline for Expanding the VDJbase Database

03/2026-Present

*Research Assistant (Supervisor: Prof. Gur Yaari, Yale University)*

New Haven, CT, U.S.

- Designed and built an end-to-end data-ingestion pipeline to transfer assembled and FASTQ-format AIRR-seq data from VDJbase files to the Yale HPC cluster via igserver; executed the nf-core/airrflow workflow using Nextflow/DSL2 and Slurm for V(D)J assignment, novel allele inference, and genotype/haplotype calling.
- Maintained reproducible pipeline execution across large-scale HPC jobs and diagnosed workflow failures, including session lock conflicts and resource allocation errors, to ensure consistent data processing.
- Supported expansion of VDJbase’s population-level germline sequence database by integrating processed repertoire data into the data submission and curation workflow.

### DrugPlayGround: Benchmarking Large Language Models and Embeddings for Drug Discovery

05/2025-01/2026

*Research Assistant (Supervisor: Prof. Hongyu Zhao, Yale University)*

New Haven, CT, U.S.

- Investigated LLM-based drug representations as an alternative to structure-based molecular embeddings, evaluating whether semantic knowledge encoded in language models captures pharmacological information beyond molecular structure.
- Benchmarked 5 LLMs (GPT, Gemini, Claude, DeepSeek, Mistral) and 5 embedding models across drug synergy prediction, drug-protein interaction, and chemical perturbation response tasks using a unified framework; curated 4 datasets (1,000+ cancer drug entries each) from PubChem and HuggingFace.
- Engineered Normal, CoT, and Meta prompting strategies; developed batch-inference pipelines on Yale HPC (Slurm) generating 100k+ drug descriptions, with domain-specific baselines (Uni-Mol, ChemCPA, RDKit) for comparison.
- Implemented evaluation metrics (BERT, ROUGE, BLEU, cosine similarity) with temperature-sensitivity analysis; produced visualizations (bubble heatmaps, box plots) to synthesize results
- Demonstrated that LLM-derived embeddings consistently outperform state-of-the-art molecular foundation models, achieving up to 54% improvement in drug synergy prediction ( $R^2$ ) and 22% improvement in drug-protein interaction accuracy.
- Co-authored a manuscript submitted to npj Drug Discovery.

## Diabetes Prevention Chatbots to Help Dietary Self-Management for the Chinese Population

06/2023-05/2025

Research Investigator (Supervisor: Prof. Ming-Chun Huang, DKU)

Kunshan, Jiangsu, China

- Investigated the use of LLMs for digital health interventions in culturally tailored diabetes prevention, focusing on dietary and lifestyle self-management support for Chinese-speaking populations.
- Synthesized 5+ academic papers to curate a China-specific diabetes prevention guideline; constructed an 80+ entry JavaScript conversational dataset grounded in dietary and lifestyle evidence.
- Fine-tuned GPT-3.5-turbo and GPT-4o via OpenAI API on the curated dataset; conducted scenario-based evaluation combining quantitative metrics (training and validation loss) with qualitative assessment (feedback examples, contextual appropriateness) for health counseling tasks.
- Developed and deployed an AI-driven Streamlit web application with Firebase authentication, real-time chat history storage, and export functionality.
- Demonstrated that fine-tuned GPT 3.5-turbo showed better generalization performance and relatively lower loss; published a first-author paper at EMBC 2025.

## SELECTED ADDITIONAL RESEARCH EXPERIENCE

1. Improving the Decision Boundary of Non-Parametric Outlier Synthesis (NPOS) Model | Supervisor: Prof. Kaizhu Huang, DKU | 01/2025-03/2025
2. Systematic Review: Key Factors in AI Algorithms Applied in Cancer Treatment | Supervisor: Prof. Meifang Chen, DKU | 06/2023-03/2024

## DATA SCIENTIST EXPERIENCE

Merit Interactive Co., Ltd.

06/2024-08/2024

Data Scientist Intern

Hangzhou, Zhejiang, China

- Processed and standardized 100M+ healthcare and pharmacy records using FlinkSQL and Python within a cloud-based data governance pipeline, enabling cross-institutional data integration and quality management workflows.
- Developed interactive multidimensional visualizations, including heatmaps, bar charts, and pie charts, to analyze medication sales, hospital operations, and market trends for strategic marketing decisions.
- Built and tested 3 customer-facing APIs for pharmaceutical dataset access; optimized their performance and reliability with Python and JavaScript and deployed improved code to GitLab for version control.

## TECHNICAL SKILLS

- **Programming Languages:** Python (TensorFlow, PyTorch, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Streamlit, LangChain), R (ggplot2, tidyverse, bootstrap), SQL, Slurm, Java, JavaScript, HTML, CSS
- **Machine Learning & AI:** Deep Learning (Neural Networks), Transformers, LLMs, RAG, LangChain, Hugging Face, Prompt Engineering, and Hyperparameter Tuning
- **Development Tools:** Git, VS Code, Android Studio, RStudio, Jupyter Notebook, Anaconda, Yale High-Performance Computing (HPC) Cluster, Cloud Computing Platforms (AWS, Azure), Jenkins, Nextflow, nf-core/airrflow
- **Data Visualization:** Tableau, Power BI, Canva